MIL-T-48825(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

TRAINING CABLE ASSEMBLY W5-F

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of trainer cable assembly entitled as Training Cable Assembly W5-F which is used with the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For.

DRAWINGS

ORDNANCE CORPS

8846048	- Cable Assembly W5-F.
8846373	- Warhead Section Assembly Packing Drawing.
8846971	- Bag, Packing, Ammunition, for Cable Assembly W5-F.
EL-8855804	- Equipment List.

FSC 1135

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cable assembly shall comply with all requirements specified on the drawing (dwg) 8846048, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Continuity. The continuity of the cable assembly shall be 0.25 ohms maximum (max) from pin to pin as specified in Table I when tested in accordance with 4.4.1.

Table I

From	To	From	To
P3-X	P1-N	P3-d	P1-V
P3-Y	P1-N	P3-e	P1-V
P3-Z	P1-P	P3-E	P1-G
P3-a	P1-P	P3-F	P1-G
P3-A	P1-X	P3-G	P1-H
P3-B	P1-X	P3-M	P1-H
P3-C	P1-W	P3-S	P1-D
P3-D	P1-W	P3-T	P1-a
P3-b	P1-E	P3-R	P1-S
P3-c	P1-E		

3.6 Insulation resistance. The insulation resistance of the cable assembly when subjected to a potential of 500 + 50 volts direct current (vdc) shall not be less than 20 megohms between the circuits specified in Table II when tested as specified in 4.4.2.

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Table II

Connector Shells	All pins	Except pins
P3-X	All other pins	P1-N and P3-X
P3-Z	All other pins	P1-P and P3-a
P3-A	All other pins	P1-X and P2-B
P3-C	All other pins	P1-W and p3-D
P3-b	All other pins	P1-E and P3-c
P3-d	All other pins	P1-V and P3-e
P3-E	All other pins	P1-G and P3-F
P3-G	All other pins	P1-H and P2-M
P3-S	All other pins	P1-D
P3-T	All other pins	P1-a
P3-R	All other pins	P1-S

3.7 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8846048) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

Table III
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity (see 3.5)		X
3. Insulation resistance (see 3.6)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, preformance storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristics table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequency for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cable assemblies, W5-F prior to potting or prior to securing connector shell and cable clamp (see dwg 8846048).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101.	Insulation sleeving torn or damaged	Visual
102.	Insulation sleeving loose	Visual/Manual
103.	Evidence of poor workmanship (see 3.8)	Visual
104.	Improperly soldered joints (see 3.7)	Visual
105.	Any operation missing	Visual
106.	Any pin bent	Visual
*107.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pins)	Visual/Gage
**108.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pins)	Visual/Gage
109.	Pin missing (as evidenced by empty hole in insert)	Visual
110.	Foreign matter on any pin or in any pin hole unless removable	Visual
111.	Crack in any component	Visual
112.	Portion of connector broken away (minor chips, scratches, burrs or gouges that do not affect the mating are not objectionable	Visual
113.	Deformation in any area	Visual
114.	Distance from connected shields to bottom of P1, P4, and P5 connectors	Visual
115.	Objective evidence relative to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.2 Cable assembly W5-F (see dwg 8846048).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Distance between P1 connector and junction of P1 and P3 cables	Gage

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102.	Distance between P1 and P3 cable junction and P3 connector	Gage
103.	Distance between junction P1, P4, and P5 cables and P4 connector	Gage
104.	Distance between junction of P2 and P6 cables and P6 connector	Gage
105.	Distance between P1 connector and junction of P1, P4, and P5 cables	Gage
106.	Distance between P1 connector and P2 connector	Gage
107.	Distance between P2 connector and junction of P2 and P6 cables	Gage
108.	Gap between connector cover and plate, min	Gage
*109.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pins)	Visual/Gage
**110.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in height relative to adjacent pins)	Visual/Gage
111.	Any components damged	Visual
112.	Printed tape missing	Visual
113.	Marking misleading, unidentifiable or mislocated	Visual
114.	Evidence of poor workmanship (see 3.8)	Visual
115.	Any components missing or loose	Visual/Manual
116.	"Training Use Only" tape missing	Visual
117.	Objective evidence relating to material missing or improper (see 3.1)	Visual
118.	Serialization missing or improper (see 3.4)	Visual

Minor: None defined.

4.3.2.1.3 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation and packaging not in accordance with MIL-STD-1169	Visual
102.	Packing not in accordance with dwgs 8846373 and 8846971 as applicable	Visual

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103. Marking not in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.8).....Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4.1) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent socket, Gage (see 4.3.4.1) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.063 inch max. from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector insert shall be utilized as the reference surface for all measurements of socket depths.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity	Major
b. Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity and insulation resistance. Each cable in the lot shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL (see EL-8855804).

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When either of the conditions listed below exists, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity. The circuit continuity of the W5-F cable shall be tested utilizing a current of 100 ± 1 ma and an open circuit voltage of 5 ± 0.5 vdc maximum.

4.4.2 Insulation resistance. The insulation resistance of the W5-F cable shall be measured utilizing a 500 ± 50 volts potential dc. The current utilized shall be limited to 5 ma (max). Observe each reading in megohms.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The cable shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Shipment with trainer. When shipped as a component of the trainer, the cables shall be installed in accordance with dwg 8846373.

5.2.2 Spares.

5.2.2.1 Levels A and C. The cables when shipped as spares shall be packed in accordance with dwg 8846971.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2122 dated 5 December 1963.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A194

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